HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Information and communications technology

Software Requirement Specification

Version 1.2

AIMS Project

Subject: ITSS Software Development

Group 16
Nguyễn Huy Diễn – 20235910
Phạm Đức Phước – 20235988
Nguyễn Tiến Đạt – 20239715
Lê Hoàng Kiên – 20235958

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1 Introduction

1.1 Objective

The objective of this Software Requirement Specification document is to provide a comprehensive requirement description of the AIMS system. Specifically, the document aims to provide a clear definition of functional and non-functional requirements, system behaviors, and constraints for the system.

This document acts as a communication bridge between the stakeholders involved in the development of the platform, serving as the authoritative shared agreement and understanding of the system's capabilities and expected outcome. Intended audience of the document consists of:

- System developers: who will implement features of the system
- System designers: who will design the system architecture and components
- Testers: who will verify that the system satisfies the requirements
- Project instructors and reviewers: who will evaluate system specifications

1.2 Scope

The software product to be described in this document is called AIMS (An Internet Media Store), an e-commerce software that supports the buying and selling of physical media products. The system supports several types of physical media products including Books, Newspapers, CDs, and DVDs. Customers can browse for products, add products to a virtual shopping cart, and place orders for their purchases. Besides, the system also provides administrative features for users to manage products, orders, and users. In particular, the Product Manager can add, update, remove products, or manage stock, whereas the Administrator can manage user accounts and system roles.

The system aims to provide a convenient platform for purchasing media products while ensuring efficient product management and order processing. It also integrates payment services to support secure transaction processing.

1.3 Glossary

No	*Term*	*Explanation*	*Example*	*Note*
1	AIMS	Stands for "An Internet Media Store", an e-commerce software that supports the buying and selling of physical media products		The core system described in this document

No	***Term***	***Explanation***	***Example***	***Note***
2	Physical media	The physical products available for retail on the AIMS platform	Books, Newspapers, CDs, and DVDs	The architecture allows for future inclusion of new product categories
3	VAT	Value Added Tax. The system strictly applies a 10% VAT for all physical media items		
4	PayPal Sandbox	A third-party service provider (external actor) integrated with AIMS to facilitate financial transactions via credit card processing		Used for validating transaction tokens, processing payments, and handling refunds
5	VietQR	A third-party service provider (external actor) integrated with AIMS to process QR-based transfer payments	VietQR Test Callback API, Transaction Sync API	Generates different types of QR codes (Dynamic, Static, Semi-dynamic) Must be unique and contain exactly 8 digits
6	API	Application Programming Interface. Used by AIMS to communicate securely with third-party payment systems like VietQR and PayPal		
7	CUD	Stands for Create, Update, Delete. Represents the lifecycle management actions a Product manager can perform on the product catalog		
8	ISSN	An 8-digit unique number used as an identifier specifically for newspaper type products		
9	Barcode	A unique identifier for products, preventing duplication in the system and not including special characters	202312345	

No	Term	Explanation	Example	Note
10	Token	A temporary authentication code that allows the AIMS system to access VietQR APIs. The token perform payment-related requests.		The token is valid for 300 seconds and must be requested again after it expires.

1.4 References

- Problem Statement for AIMS Software System: AIMS-ProblemStatement-ver3.1.1
- **[TEMPLATE-EN] Software Requirements Specification (SRS) v1.2**, Course Template Document
- **Software Requirements Specification (SRS)**,Software Engineering Course Materials, Hanoi University of Science and Technology.
- **PayPal Sandbox (Credit Card Payment / Refund)**
  https://developer.paypal.com/docs/api/payments/v2
  https://developer.paypal.com/api/rest/
  https://developer.paypal.com/tools/sandbox/
- **VietQR (QR Code Payment) documentation**
  https://api.vietqr.vn/en
  https://api.vietqr.vn/en/vn/overview/description-of-vietqr-api-workflow
  https://doc.vietqr.vn/vietqr-doc/api-vietqr-callback/api-vietqr-host2host/integrated-document-for-payment-service-vietqr

2 Overall Description

2.1 Survey

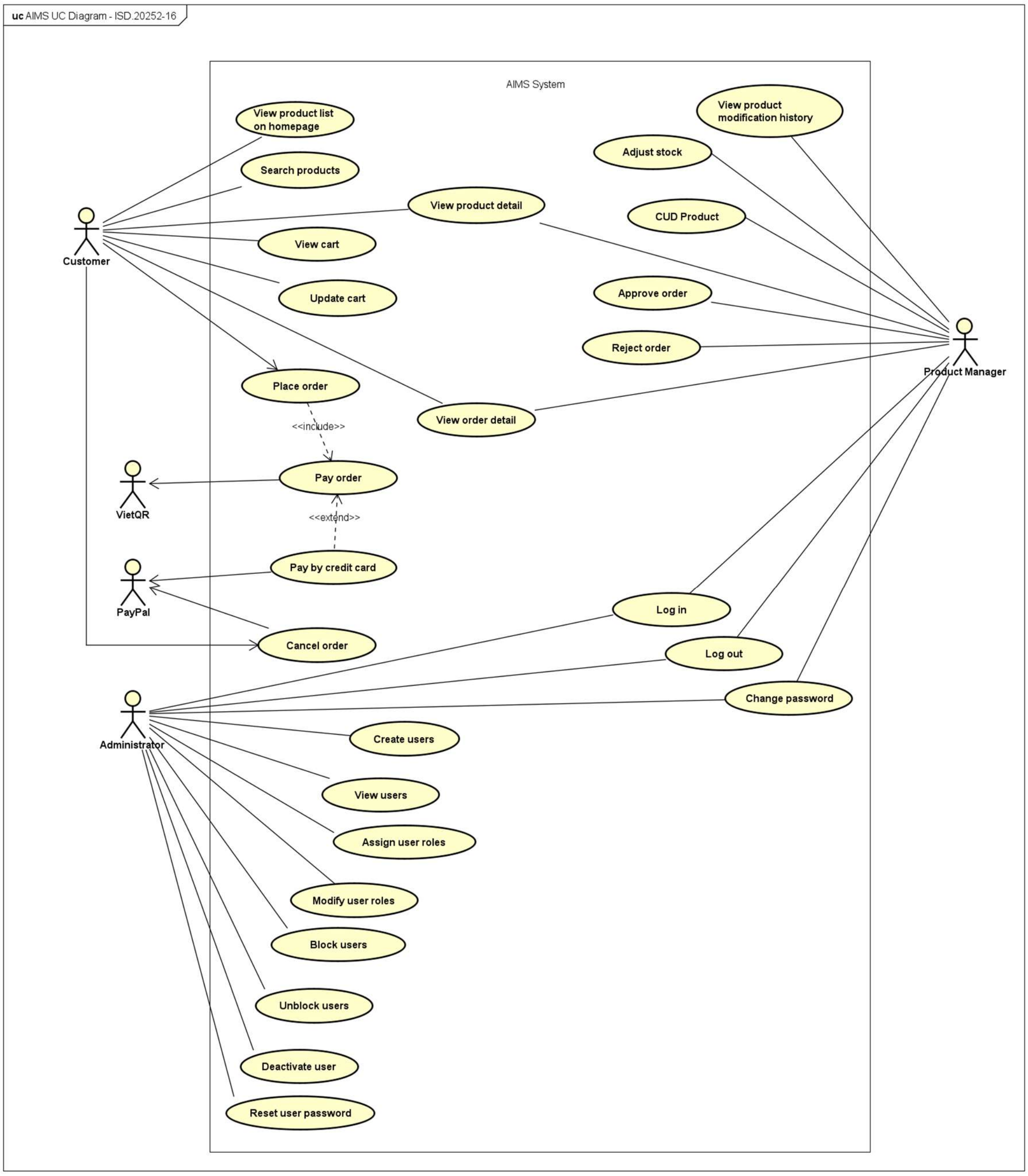
AIMS is a specialized e-commerce platform designed for the retail of physical media products, including Books, Newspapers, CDs, and DVDs. The system is designed to provide a seamless, 24/7 shopping experience, that prioritizes high availability and user-friendliness for both existing and first-time users.

From a technical perspective, the software is built to handle high-traffic demands, supporting up to 1,000 concurrent users while maintaining a rapid response time of under 2 seconds during normal operations. AIMS serves as a comprehensive management and sales hub, integrating inventory control, product catalog maintenance, and secure electronic payment processing. The system's architecture is designed with extensibility in mind, allowing for the future inclusion of new product categories beyond the current physical media scope.

The following actors interact with the AIMS system to fulfill various business and operational requirements:

- Customer: The primary end-user of the system. Customers interact with the software to browse the product catalog, search for items, and manage their shopping carts. They can proceed through the checkout process and perform payments without a mandatory initial login.
- Product manager: An internal staff member responsible for the lifecycle of the store's inventory. This actor performs crucial administrative tasks such as adding new media to the stock, updating existing product details, and managing stock levels. The Product manager is also restricted by specific system constraints, such as daily deletion limits, to ensure data integrity and prevent unauthorized mass removal of items.
- Administrator: The privileged user in charge of system security and user management. The Administrator handles the creation and maintenance of user accounts, assigns functional roles, and manages access control by locking or unlocking accounts as necessary.
- Payment System (External Actor): A third-party service provider that integrates with AIMS to facilitate financial transactions. This includes PayPal Sandbox for credit card processing and VietQR for QR-based transfers. This actor is responsible for validating transaction tokens, processing payments, and handling refund requests through secure API communication.

2.2 Overall requirements



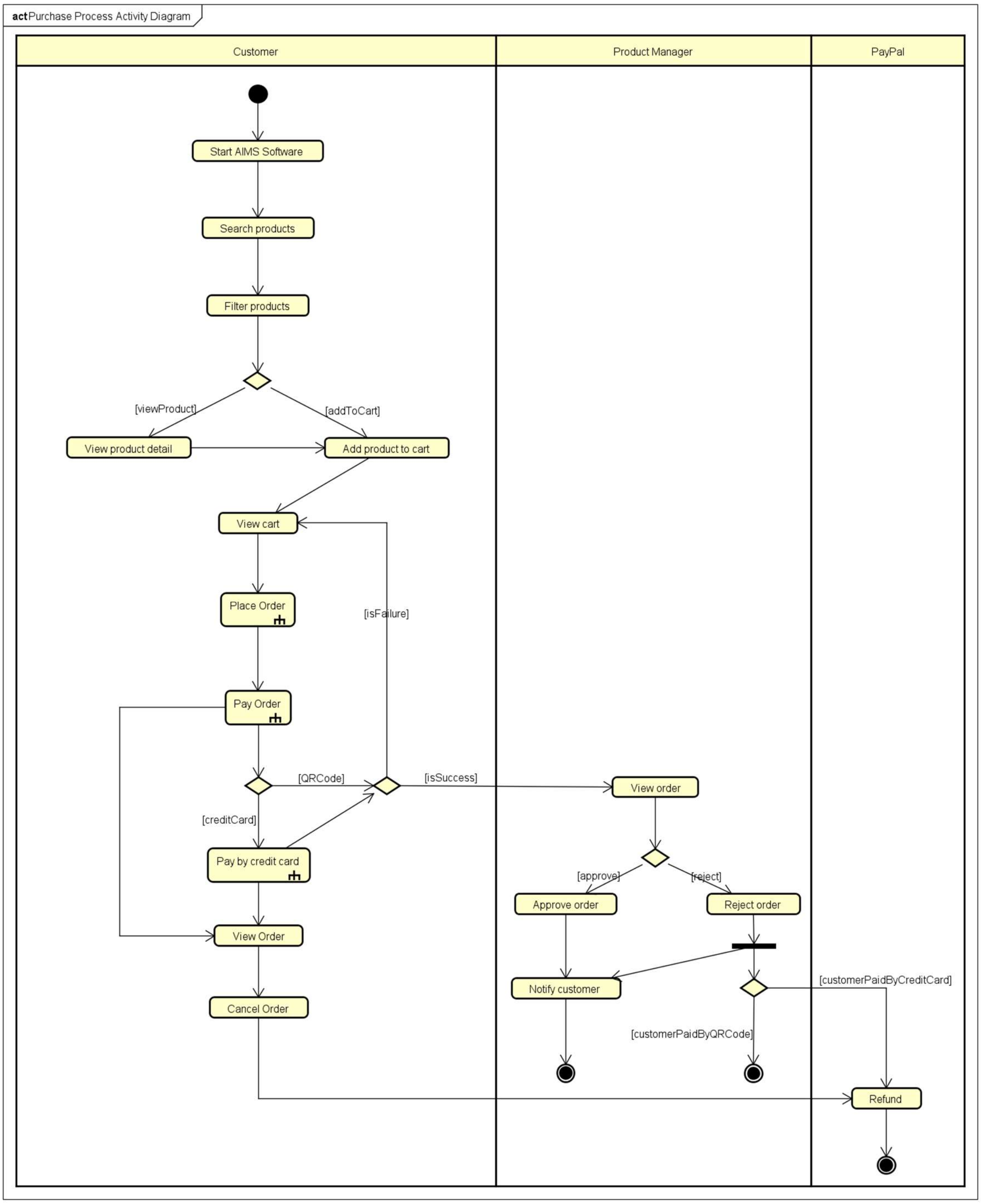
AIMS Use Case Diagram

2.3 Business process

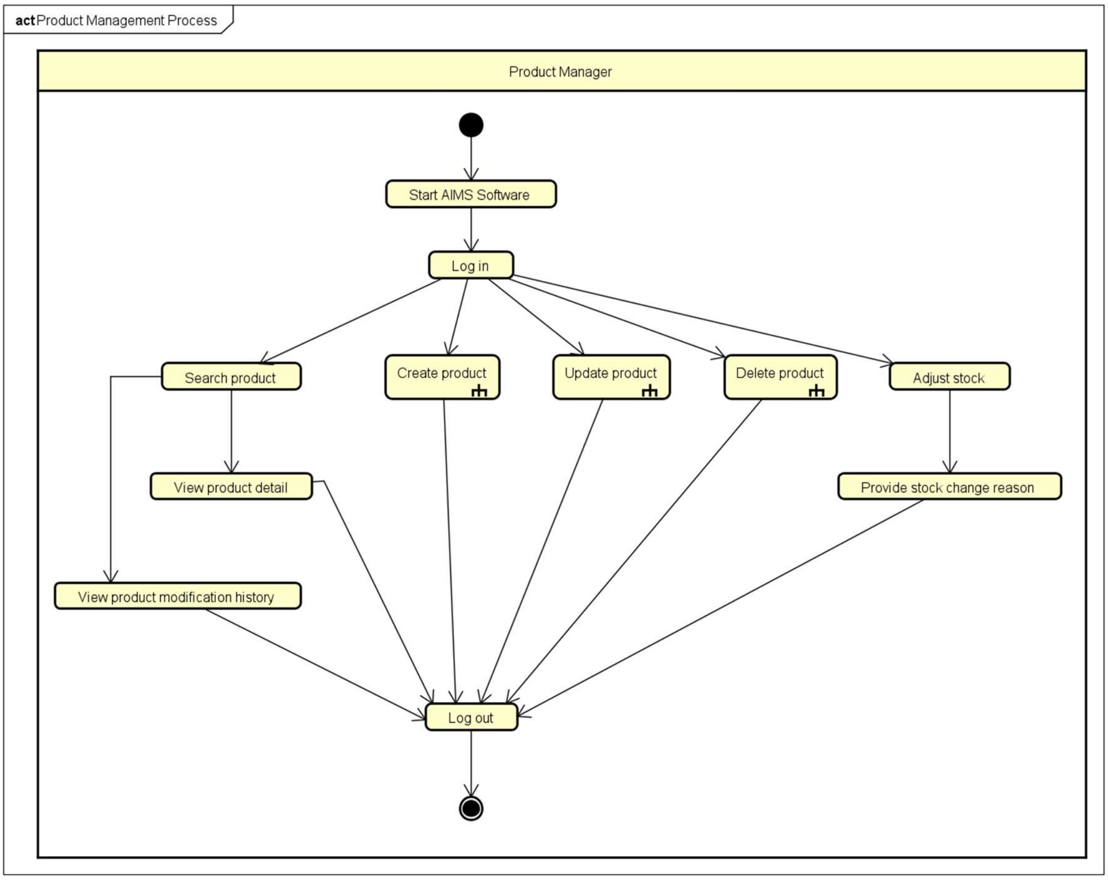
The AIMS System supports three core business processes representing the main functionalities of the system. These processes describe how the system operates to support product buying and selling, product management, and system administration. The three primary business processes identified in the system are:

- Purchase Process: which describes how orders are created and processed from product selection to payment and order confirmation.
- Product Management Process: which describes how product information and inventory are maintained within the system.
- Administration Process: which describes how user accounts and system access are managed.

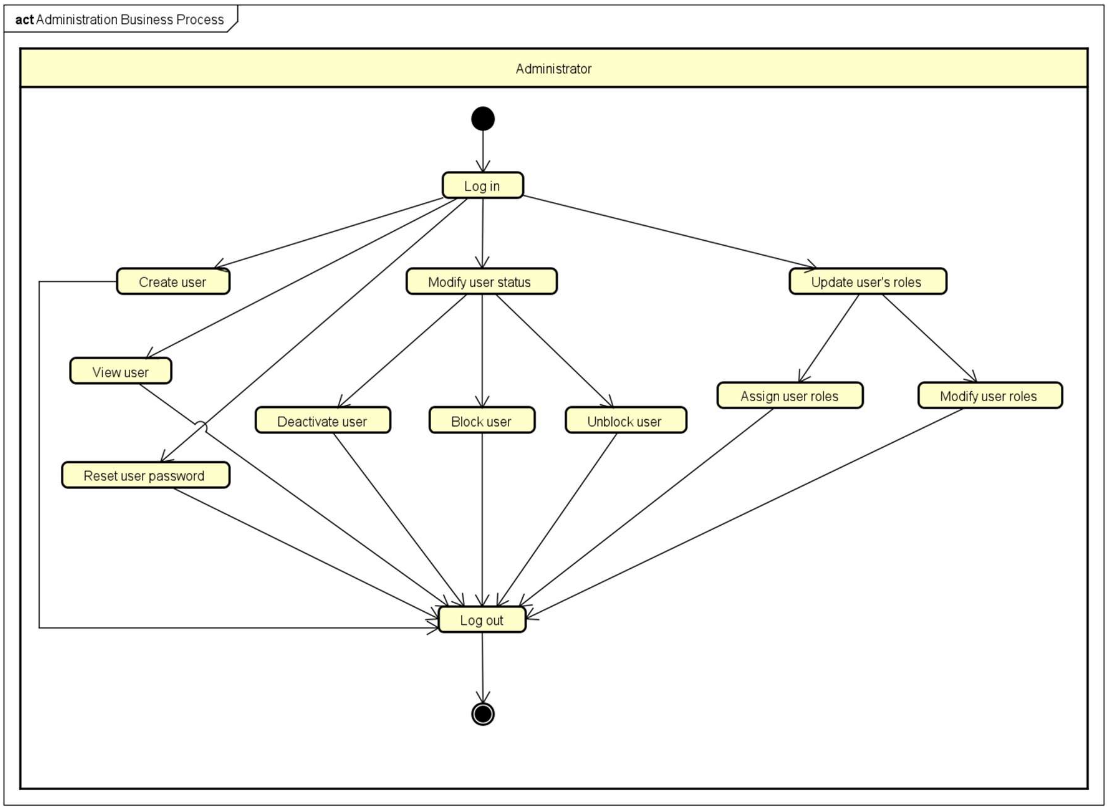
The activity diagrams below illustrate the workflows associated with these three business processes. Each diagram presents the sequence of related use cases and the interactions between the actors during the execution of these processes.



Purchase Process Activity Diagram



Product Management Process Activity Diagram



Administration Process Activity Diagram

3 Detailed Requirements

3.1 Use case UC001 – Place Order

Use Case Place Order

1. **Use Case Code**

UC001

2. **Brief Description**

The use case "Place Order" describes the interaction between the customer and the AIMS Software. It is invoked when the customer wishes to place an order after the Customer has finished selecting products.

3. **Actor(s)**

There is only 1 Actor interacting with this use case: Customer.

4. **Precondition(s)**

The use case can only be invoked if the following condition is satisfied:

- Customer has added at least 1 product in cart.

5. **Basic Flow of Events**

Step 1. Customer places order

Step 2. AIMS Software checks for stock availability

Step 3. AIMS Software displays the delivery form and prompts the customer to set up delivery information

Step 4. Customer enters delivery information and submits

Step 5. AIMS Software checks for missing required fields and input validity

Step 6. AIMS Software calculates and displays the delivery fee

Step 7. AIMS Software displays and temporarily saves invoice information

Step 8. Customer requests to pay for the order

Step 9. AIMS Software calls the use case "Pay Order"

Step 10. AIMS Software puts the order in pending state

Step 11. AIMS Software displays general order information and transaction information to screen

Step 12. AIMS Software sends invoice and payment transaction information to customer's email

Step 13. AIMS Software records payment transaction information and successfully paid order

6. Alternative Flows

No	Location	Condition	Action	Resume Location
1	Step 2	Stock is insufficient	AIMS software prompts the customer to update the cart and displays the available quantity for each affected product.	Use case ends
2	Step 5	Missing required fields or invalid inputs	AIMS Software prompts the customer to fill in missing fields or reenter	Step 4
3	Step 7	Customer requests to adjust the delivery method	AIMS Software displays the delivery form and prompts the customer to reenter	Step 4
4	Step 7	Customer requests to adjust product list	AIMS Software displays cart and prompts the customer to update the cart	Step 1
5	Any step	Customer requests to go back to step N	AIMS Software redirects to step N	Step N
6	Any step	Customer exits the software or requests to cancel order	AIMS Software cancels the ordering process	Use case ends
7	Step 9	Order payment is not successful	AIMS Software displays notification of failed transaction	Use case ends

Table 1-Alternative flows of events for UC Place Order

7. Input data

No	Data Fields	Description	Mandatory	Valid Condition	Example
1	Customer name	Name of the customer/receiver	Yes		Nguyễn Văn A

2	Phone number	Contact number of the customer that will be used to contact the customer	Yes	- 10 digits - Starts with 0 - May contain separators such as '-', '.', or space in between digits	0123456789 0123.456.789 0123-456-789 0123 456 789 012.34.56.789 …
3	Province/City (Tỉnh/Thành phố)	A province/city in Viet Nam, selectable from a list	Yes		Hà Nội Nghệ An …
4	Ward/Commune (Phường/Xã)	A ward/commune in that province/city, selectable from a list	Yes		Phường Hà Đông Xã An Khánh …
5	Address	Detailed address of the customer attached to	Yes		199 Chiến Thắng, Triều Khúc
6	Delivery method	How the products should be delivered to the customer	Yes		Quick, Normal (default), …

Table 2-Input data of Delivery Information

8. Output data

No	Data Fields	Description	Display Format	Example
1	Product type + title	The type and the title of the product	[*Type*] [Title]	*Book* Data Structures & Algorithms *DVD* Ice Age Collection …

2	Quantity	The number of the corresponding product to be purchased	Positive integer	3
3	Price	Price of the product	Positive integer Includes comma for thousands	99,000
4	Amount	Total price for the product	Positive integer Includes comma for thousands	297,000
5	Total product price	Total amount of all products	Positive integer Includes comma for thousands	1,423,000
6	Total product price with VAT	Total product price + VAT	Positive integer Includes comma for thousands	1,585,300
7	Delivery fee		Positive integer Includes comma for thousands	50,000
8	Total amount	The amount to be paid	Positive integer Includes comma for thousands	1,635,300

Table 3-Output data of Invoice Information

No	Data Fields	Description	Display Format	Example
1	Customer name			Nguyễn Văn A
2	Phone number		10 digits starting with 0 with '.', '-', or space	0123456789 0123.456.789 0123-456-789 0123 456 789 012.34.56.789 …
3	Province/City			Hà Nội

	(Tỉnh/Thành phố)			Nghệ An …
4	Ward/Commune (Phường/Xã)			Phường Hà Đông Xã An Khánh …
5	Address			199 Chiến Thắng, Triều Khúc
6	Delivery instructions			
7	Total amount	The amount to be paid	Positive integer Includes comma for thousands	1,635,300

Table 4-Output data of Order Information

No	Data Fields	Description	Display Format	Example
1	Transaction ID			
2	Transaction Content			
3	Transaction Datetime		dd/mm/yyyy	15/03/2026

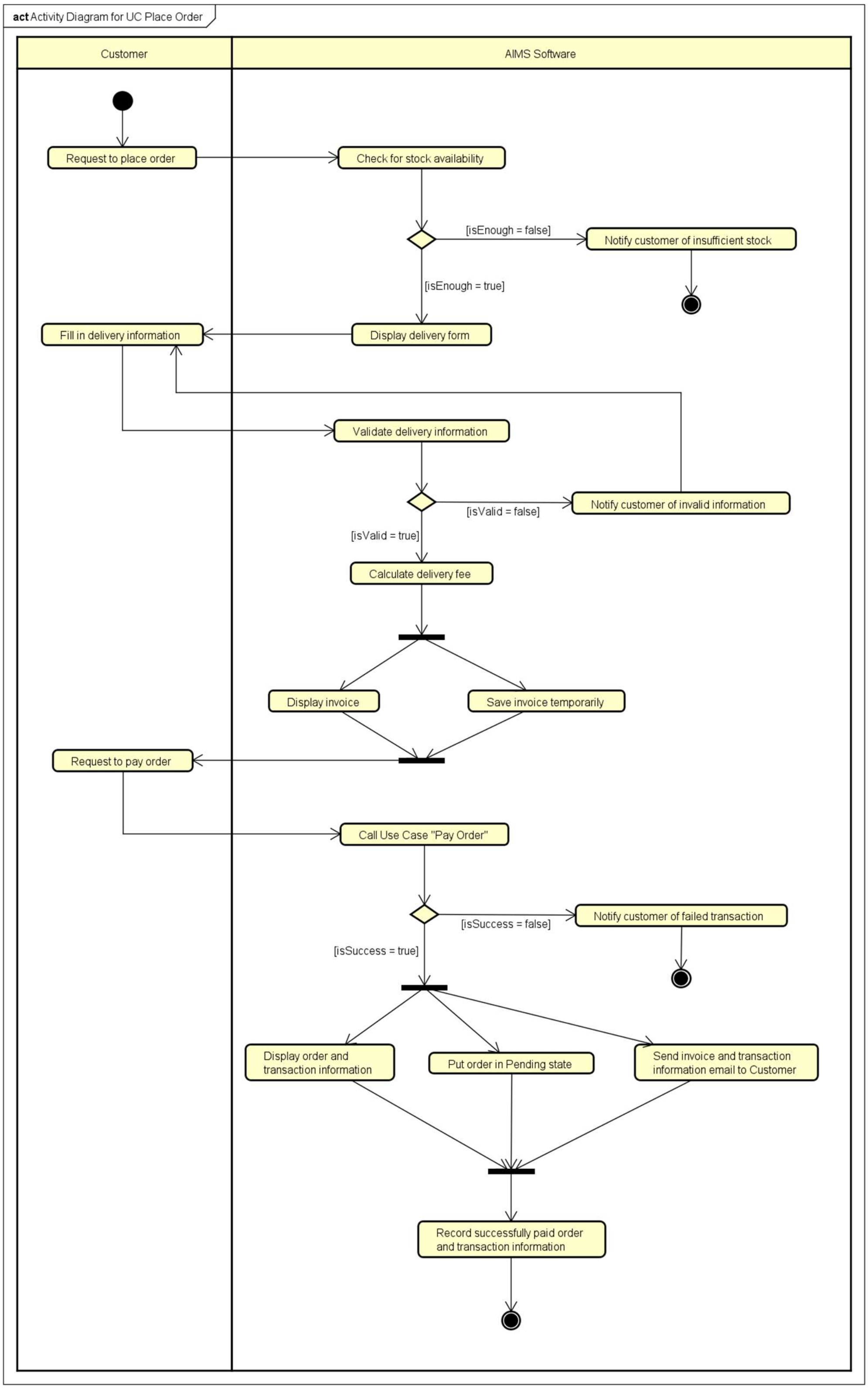
Table 5-Output data of Transaction Information

9. Postconditions

The use case can be considered successful if all the following conditions are satisfied:

- A new order is created in the system and its status is set to Pending.
- The system records payment transaction information and the successfully paid order
- Customer receives email containing invoice and payment transaction information

If the use case ends prematurely or is unsuccessful due to payment failure, the ordering process is reset and nothing else happens.



UC001 Place Order Activity Diagram

3.2 Use case UC002 – Pay Order

Use Case "Pay Order"

1. Use case code

UC002

2. Brief Description

This use case describes the interaction between AIMS software with the customer and VietQR when the customer wishes to pay the order

3. Actors

3.1 Customer

3.2 VietQr

4. Preconditions

Customer has entered valid delivery information

Aims software displayed the temporary invoice information for the order

5. Basic Flow of Events

1. Customer requests to pay order
2. AIMS software calls the get token API from VietQR
3. VietQR sends the access token to AIMS software
4. AIMS software calls the API of VietQR to generate the QR code (see Table 1)
5. VietQR sends the QR code for AIMS software
6. AIMS software displays the generated QR code along with the invoice information on the payment screen (see Table 2)
7. Customer triggers the Test Callback API
8. AIMS software calls the VietQR Test Callback API to simulate a successful payment transaction (see Table 3)
9. VietQR sends a payment callback to AIMS software
10. AIMS software verifies the callback information
11. AIMS software confirms the payment result and updates the order status

6. Alternative flows

No	Location	Condition	Action	Resume location
1.	At Step 1	If the customer choose to pay order by credit card	AIMS Software inserts the use case "Pay order by credit card"	Use case ends

2.	At Step 6	if the customer wants to cancel the payment	AIMS Software navigates the customer back to the Invoice Screen and no order or payment made	Use case ends
3.	At Step 9	If the payment callback indicates a failed transaction	Aims software notifies the customer that the transaction has failed	Resumes at Step 6

7. Input data

Table 1-Input data of QR code generation API (VietQR system)

No	Data fields	Description	Mandatory	Valid condition	Example
1.	BankCode	Code of the receiving bank account	Yes	Valid bank code supported by VietQR	MB
2.	BankAccount	Bank account number for the transfer	Yes	Valid bank account number	113000045678
3.	UserBankName	Name of the bank account holder	Yes	Must not contain Vietnamese accents	Nguyen Dang Quang
4.	Content	Payment transfer content	Yes	Maximum 19 characters, no special characters	ORDER112
5.	qrType	Type of VietQR code to generate	Yes	Dynamic QR = 0, Static QR = 1, Semi-dynamic QR = 3	0
6.	Amount	Total money of the corresponding media	No	- qrType = 0 or qrType = 3 required	175000

				- Positive integer	
7.	OrderId	Unique identifier of the order	No	≤ 13 characters	ORDER001

Table 2-Input data of VietQR Test Callback API (VietQR system)

No	Data fields	Description	Mandatory	Valid condition	Example
1.	BankAccount	Bank account number for the transfer	Yes	Valid bank account number	113000045678
2.	Content	Payment transfer content	Yes		ORDER112
3.	Amount	Total money of the corresponding media	Yes	- Positive integer	300000
4.	TransType	Payment transfer content	Yes	C = credit transaction, D = debit transaction	C
5.	BankCode	Code of the receiving bank account	Yes	Valid bank code supported by VietQR	MB

8. Output data

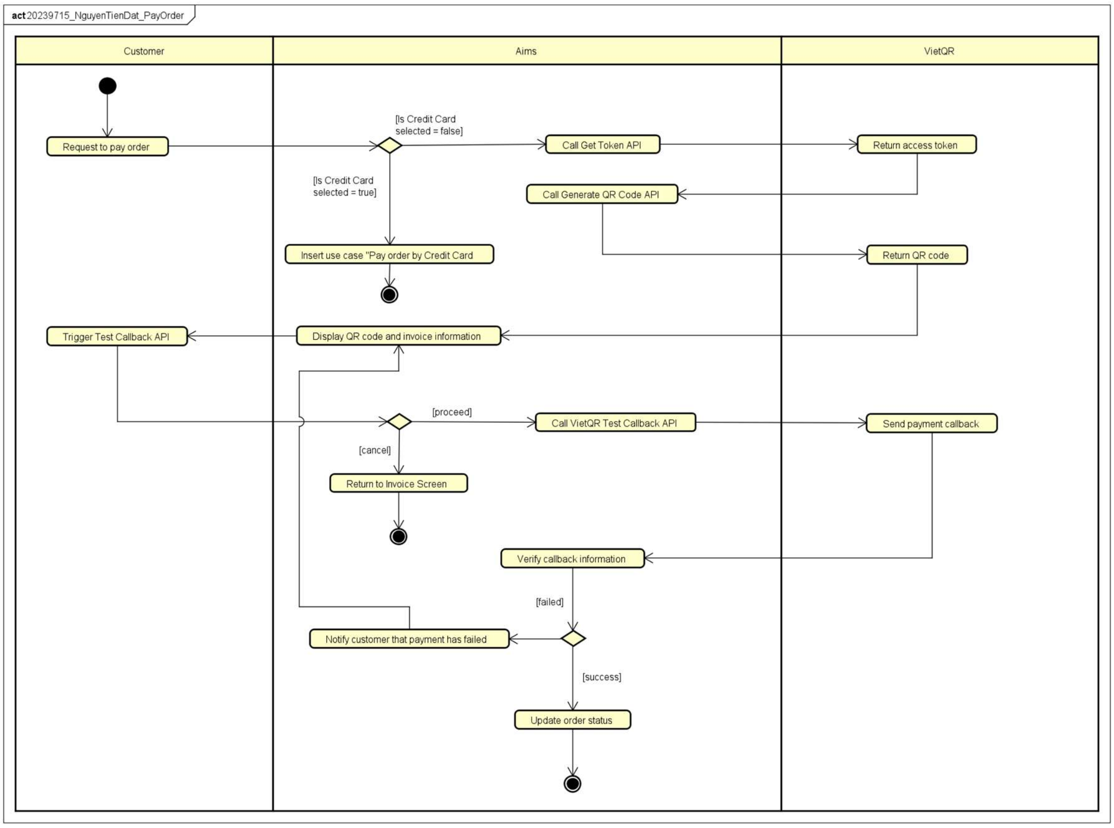
Table 3-Output data of QR code along with the invoice information (Aims software)

No	Data fields	Description	Display format	Example
1.	Bank Name	The name of the receiving bank		VietinBank
2.	Bank Account	Bank account number for the transfer		1130 0004 5678

3.	UserBankName	Name of the bank account holder		Nguyen Dang Quang
4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer	150,000 VND
5.	QR Code	QR image generated from the string	QR Image	

9. Postconditions

- After a successful payment, the order is stored in the Aims Software with the status "*Pending Processing*"
- Aims software will display general information of the order and transaction information, and then send invoice with payment transaction information to the customer's email.



UC002 Pay Order Activity Diagram

3.3 Use case UC003 – Pay Order by Credit Card

Use Case Pay Order by Credit Card

1. Use case code

UC003

2. Brief Description

The use case “Pay Order by Credit Card” describes the interaction between the Customer, the AIMS Software, and the PayPal payment gateway. It is invoked when the Customer chooses to pay for their order using a credit card rather than the default QR code method.

3. Actors

There are two actors:

Primary Actor: Customer.

Secondary Actor: PayPal (Payment Gateway).

4. Preconditions

+ The use case “Place Order” has been initiated.

+ The Customer has confirmed the invoice information.

+ The total amount, including VAT and shipping fees, has been calculated.

+ The Customer has selected "Credit Card" as the payment method.

5. Basic Flow of Events

1. AIMS Software displays the credit card information entry form.
2. Customer enters credit card details (Card Number, Holder Name, Expiration Date, and CVV) and submits.
3. AIMS Software validates the format of the entered data.
4. AIMS Software sends a payment request with the transaction amount to the PayPal system.
5. PayPal authenticates the card and checks for sufficient balance.
6. PayPal processes the transaction and returns a "Successful" status and transaction ID to AIMS Software.
7. AIMS Software records the payment transaction details.
8. AIMS Software displays general information of the order (customer name, phone number, shipping address, province, total amount) and transaction information (transaction ID, transaction content and transaction datetime).
9. AIMS Software sends invoice and payment transaction information to the customer's email.

6. Alternative flows

Table 01 - Alternative flows of events for UC Pay Order by Credit Card

No	Location	Condition	Action	Resume location
1	Step 3	Invalid card data format	AIMS Software notifies the user of the specific error	Resumes at Step 1
2	Step 5	Authentication failed or Insufficient balance	AIMS Software displays a Failed message.	Use case ends
3	Step 4	Connection timeout with PayPal	AIMS Software notifies that the payment gateway is unavailable.	Use case ends
4	Step 1	Customer cancels payment	AIMS Software returns to the payment method selection screen.	Use case ends

7. Input data

Table 02 - Input data of Credit Card

No	Data fields	Description	Mandatory	Valid condition	Example
1	Card Number	The 16-digit credit card number	Yes	Must be numeric	2560 4508 3456 6789
2	Cardholder Name	Name as it appears on the card	Yes	Alphabetic characters only	PHAM DUC PHUOC
3	Expiration Date	The card's expiry date	Yes	MM/YY format and must be in the future	03/30
4	CVV/CVC	Security code on the back	Yes	Exactly 3 digits	222

8. Output data

Table 03 - Output data of Transaction Result

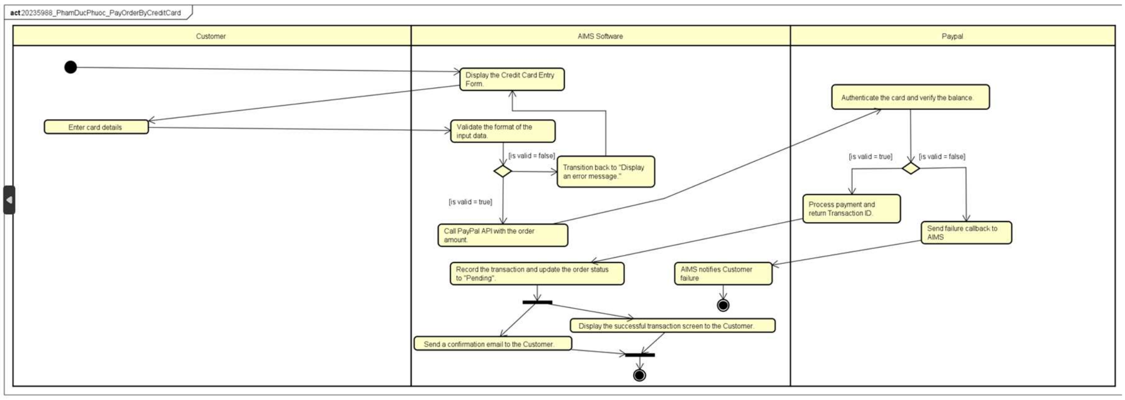
No	Data fields	Description	Display format	Example

1	Transaction ID	Unique ID provided by PayPal	Alphanumeric	PDP-12345678
2	Amount	Final charged amount	Integer with comma separator	3,636,366 VNĐ
3	Date	Date and time of transaction	dd/mm/yyyy hh:mm	03/06/2025 03:06
4	Status	Outcome of the process	Text	Successful
5	Transaction Content	Purpose of transaction	Text	Payment for order #98765

9. Postconditions

+ Success: A new transaction record is saved, the order status is updated to " Pending processing state," and the cart is emptied.

+ Failure: No payment is recorded, and the order remains unpaid.



UC003 Pay Order by Credit Card Activity Diagram

3.4 Use case UC004 – CUD Product

Use Case "CUD Product"

1. Use case code

UC004

2. Brief Description

This use case allows the Product manager to maintain the product catalog. It includes adding new physical media (Books, CDs, DVDs, Newspapers), updating existing product information, and removing products from the system according to stock and quantity constraints.

3. Actors

Product manager

4. Preconditions

The use case can only be invoked if the following conditions are satisfied:

- Product manager is logged into the AIMS software.
- The software is displaying the product management dashboard.

5. Basic Flow of Events

5.1. Basic flow: Create product

Step 1: Product manager selects the "Add product" option.

Step 2: AIMS software displays the product entry form (Title, Category, Price, Weight, etc.).

Step 3: Product manager fills in the form and selects the product type (Book, CD, DVD, or Newspaper). (see Table 2)

Step 4: AIMS software adds specific fields for that type. (see Table 3, 4, 5, 6)

Step 5: Product manager enters all required information and clicks "Save".

Step 6: AIMS software validates the data and uniqueness of the barcode.

Step 7: AIMS software saves the new product and displays a success message.

5.2. Basic flow: Update product

Step 1: Product manager selects a product from the list and clicks "Edit".

Step 2: AIMS software retrieves and displays the current information of the product. (see Table 7 and Table 9, 10, 11, 12, 13)

Step 3: Product manager modifies the desired fields.

Step 4: Product manager clicks "Update".

Step 5: AIMS software validates the changes.

Step 6: AIMS software saves updated information and notifies the Product manager.

5.3. Basic flow: Delete product

Step 1: Product manager selects one or more products (up to 10) from the list. (see Table 8)

Step 2: Product manager clicks "Delete".

Step 3: AIMS software checks the stock level for each selected product.

Step 4: AIMS software deletes selected products.

Step 5: AIMS software displays a summary of the action (number of products deleted).

6. Alternative flows

No	Location	Condition	Action	Resume location
1	At step 6 (Create) or step 5 (Update)	AIMS software detects invalid formats (e.g., negative price, empty title).	Action 1: AIMS software highlights the error fields. Action 2: The Product manager corrects the data and re-submits.	At step 7 (Create) or step 6 (Update)
2	At step 1 (Delete)	Product manager selects more than 10 products.	AIMS software prevents the action and displays failure message.	At step 1(Delete)
3	At step 2 (Delete)	Total deleted products of the day have already been 20.	AIMS software prompts a failure message of exceeding daily quota (20)	Use case ends

4	At step 2 (Delete)	Total deleted products of the day will exceed 20 after the process	AIMS software prevents the action and require to reduce the product to be deleted	At step 1 (Delete)
5	At step 3 (Delete)	Exist products in the deletion list that have stock > 0	AIMS software deactivates those products and includes number of deactivated products in the summary	At step 5 (Delete)
6	At any step	Product manager wants to cancel the operation	AIMS software return to manager dashboard	Use case ends

Table 1: Alternative flows of events for UC CUD product

7. Input data

No	Data fields	Description	Mandatory	Valid condition
1	Barcode		Yes	must be unique, not including special characters
2	Title		Yes	
3	Category	Choose from a list	Yes	
4	Original value		Yes	must be positive
5	Current value		Yes	must be positive
6	Description		No	
7	Height		Yes	must be positive
8	Width		Yes	must be positive
9	Length		Yes	must be positive
10	Weight		Yes	must be positive

Table 2: General data for UC Create product

No	Data fields	Description	Mandatory	Valid condition

1	Author		Yes	
2	Cover type	Choose from a list	Yes	
3	Publisher		Yes	
4	Publish date		Yes	
5	Number of pages		No	integer number, must be positive
6	Language		No	
7	Genre		No	

Table 3: Data for book type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Chief editor		Yes	
2	Publisher		Yes	
3	Publish date		Yes	
4	Issue number		No	
5	Publication frequency		No	
6	Language		No	
7	ISSN	an 8-digit number	No	must be unique, contains 8 digits
8	Language		No	
9	Section		No	

Table 4: Data for newspapers type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Artist		Yes	
2	Record labels		Yes	

3	Tracks list		Yes	
4	Genre		Yes	
5	Release date		No	

Table 5: Data for CD type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Disc type	Choose from a list	Yes	
2	Director		Yes	
3	Runtime		Yes	integer number, must be positive
4	Studio		Yes	
5	Language		Yes	
6	Subtitles		Yes	
7	Release date		No	
8	Genre		No	

Table 6: Data for DVD type products creation

No	Data fields	Description	Mandatory	Valid condition
1	Barcode/ID		Yes	unique

2	Data field to be modified (similar to data meant for creating product)		Yes	depend on the data to be modified

Table 7: Data for UC Update product

No	Data fields	Description	Mandatory	Valid condition
1	List of product barcode/ID that will be deleted		Yes	

Table 8: Data for UC Delete product

8. Output data

No	Data fields	Description	Display format	Example
1	Barcode	ID of the product		202312345
2	Title	Title of a media product		DVD Phim Vượt ngục
3	Category	Type of product		DVD
4	Original value	Original price of the corresponding media product	Comma for thousands separator Positive integer Right alignment	123,000

5	Current value	Current price of the corresponding media product	Comma for thousands separator Positive integer Right alignment	130,000
6	Description		Right alignment	
7	Height		Positive integer Right alignment	3 cm
8	Width		Positive integer Right alignment	10 cm
9	Length		Positive integer Right alignment	13 cm
10	Weight		Positive integer Right alignment	20 gr

Table 9: Output of general data for UC Update product

No	Data fields	Description	Display format	Example
1	Author			Nguyễn Nhật Ánh
2	Cover type			paperback
3	Publisher			Nhà xuất bản Trẻ
4	Publish date		dd/mm/yyyy	20/10/2015

Table 10: Output of data for book type products update

No	Data fields	Description	Display format	Example
1	Chief editor			Đỗ Huy Hoàng
2	Publisher			Báo Tiền Phong
3	Publish date		dd/mm/yyyy	15/3/2026

Table 11: Output of data for newspapers type products update

No	Data fields	Description	Display format	Example
1	Artist			Soobin
2	Record labels			SS Label
3	Tracks list			
4	Genre			Pop

Table 12: Output of data for CD type products update

No	Data fields	Description	Display format	Example
1	Disc type			Blu-ray
2	Director			
3	Runtime		Positive integer Right alignment	73 mins
4	Studio			
5	Language			
6	Subtitles			

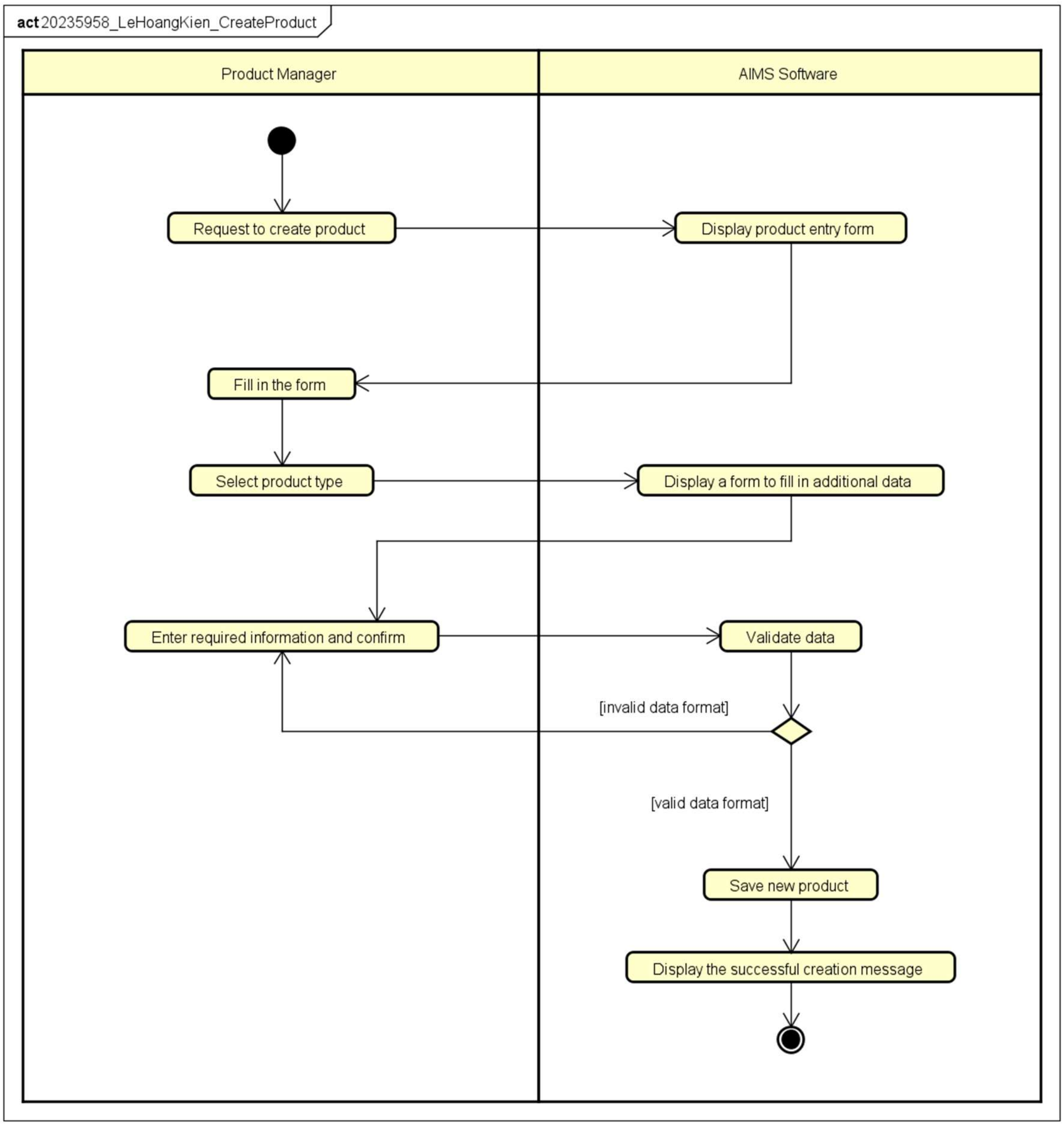
Table 13: Output of data for DVD type products update

9. Postconditions

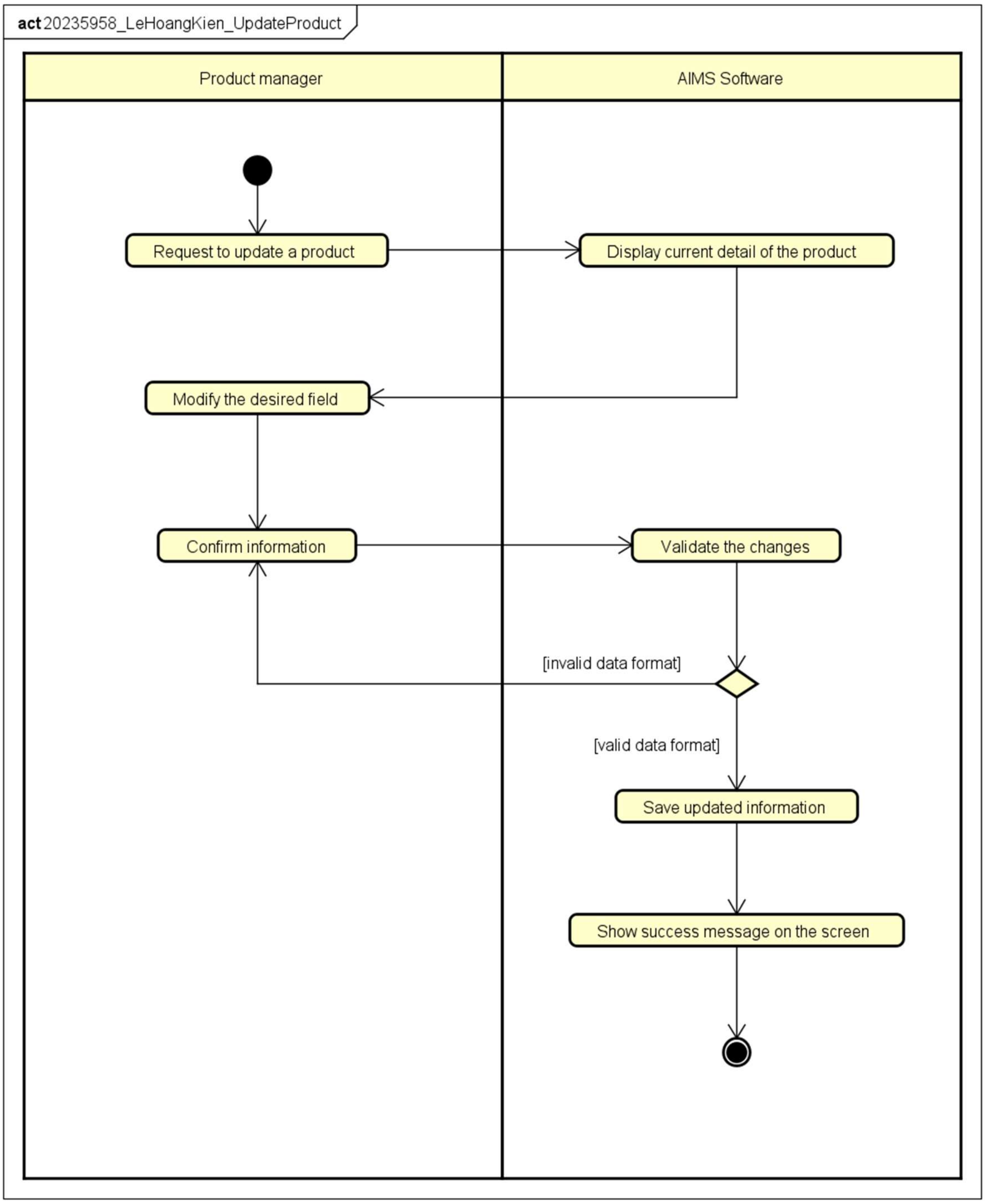
The use case is considered successful if one of the scenarios is satisfied

- Successful Create/Update: The product database is updated, and changes are reflected in the system.
- Successful Delete: Products are either removed or marked as "Deactivated," and the action is logged.

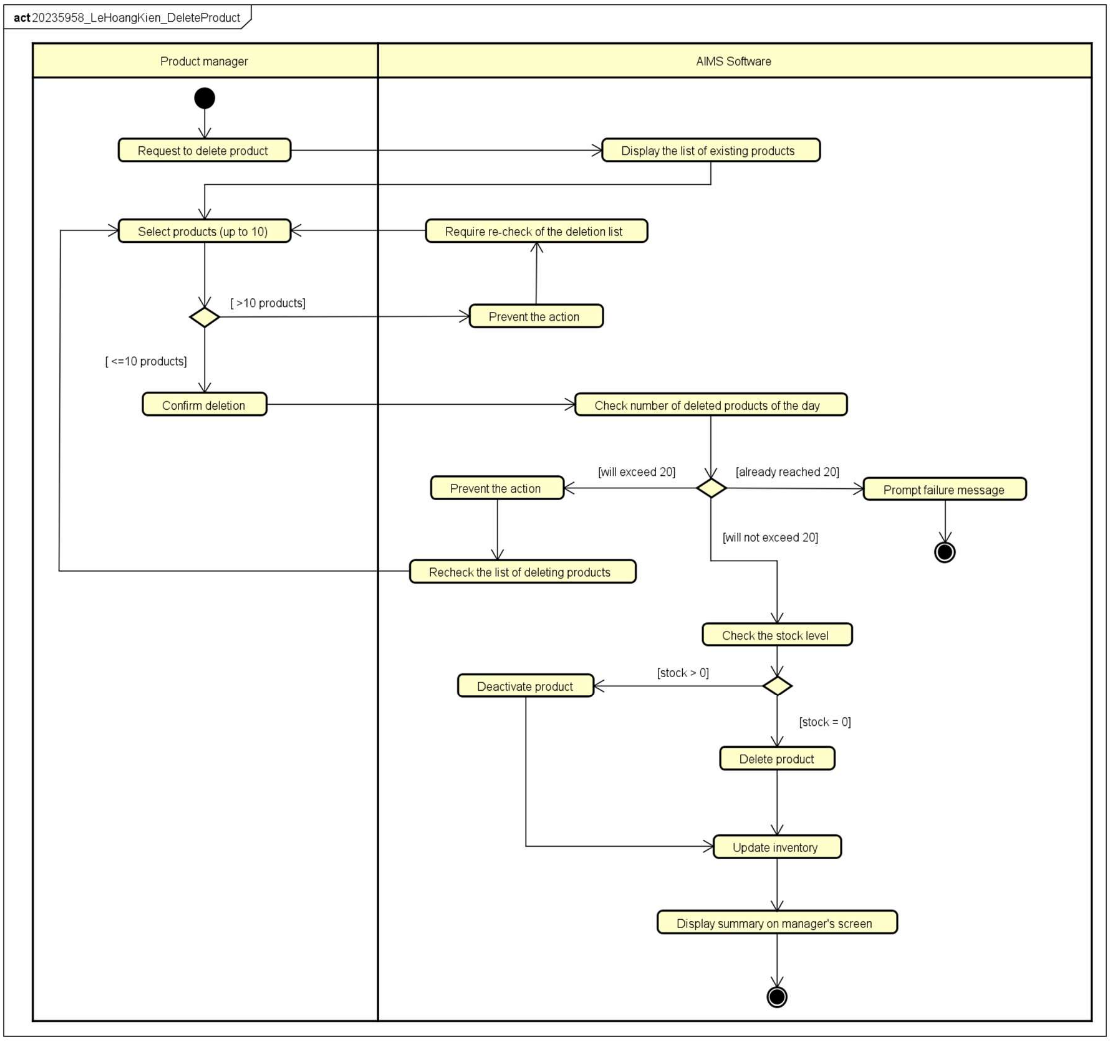
If the use case ends prematurely, the operation is reset, the software should then return to the dashboard and nothing else happens.



UC004 Create Product Activity Diagram



UC004 Update Product Activity Diagram



UC004 Delete Product Activity Diagram

4 Supplementary specification

4.1 Functionality

- The system shall log every sensitive administrative action and notify affected users. Each log captures the User ID of the performer, the timestamp, the type of operation, and a reference to the affected data.
- The system shall store the history of product creations, updates, deletions. The system shall allow Product Manager to query these histories through the system interface.
- The system shall securely hash every password so that Administrators cannot access plaintext passwords.
- The system shall validate all user inputs to ensure required fields are not empty and data matches expected formats.
- The system shall strictly apply the business rule that any current price modification must remain within the range of 30% to 150% of the product's original value.
- The system shall ensure that every product barcode is unique across the entire database.
- The system shall automatically calculate and append a 10% VAT to the current price of all media products.
- The system shall calculate shipping costs based on the product's physical attributes (weight, height, width, length) and the delivery destination.
- The system shall provide feedback (success messages, warning dialogs, or error alerts) for every user action.

4.2 Usability

- The system shall provide easy-to-use navigation to help users accomplish their tasks
- The system shall not require users to create an account, providing an easy way to browse and purchase products
- The system shall display product information in a clear and organized manner so that customers can quickly understand product characteristics such as title, category, price, and description

4.3 Reliability

- The system shall operate 24/7, providing its services at any time without interruption
- The system shall resume normal operations within a maximum of 1 hour after an incident
- The system shall maintain stable operation for at least 300 consecutive hours without failure

4.4 Performance

- The system shall support up to 1,000 customers without a significant drop in performance
- The system shall have a response time of 2 seconds under normal conditions or 5 seconds during peak hours

4.5 Supportability

- The system architecture shall be designed to allow easy addition of new media types, notification methods, and payment methods
- The system shall record system logs for important operations such as product updates, order processing, and user management activities.

4.6 Other requirements

- The system shall not allow the product manager to delete more than 20 products per day
- The system shall strictly apply a 10% VAT for all physical media items
- The system should allow additional payment methods to be integrated into the system in future versions